

➤ **Question # 01:**

```
Import java.util.Scanner;
Public class ArrayOperations {
    Static int[] array = new int[20];
    Static int size = 0;
    // 1. Add value at end
    Public static void add(Scanner sc) {

        If (size == 20) {
            System.out.println("Array is full!");
            Return;
        }
        System.out.print("Enter value: ");
        Int value = sc.nextInt();
        Array[size] = value;
        Size++;
        System.out.println("Value added successfully.");
    }
    // 2. Insert at index
    Public static void insert(Scanner sc) {
        If (size == 20) {
            System.out.println("Array is full!");
            Return;
        }
        System.out.print("Enter index: ");
        Int index = sc.nextInt();
        If (index < 0 || index > size) {
            System.out.println("Invalid index!");
            Return;
        }
    }
}
```

```

    }
    System.out.print("Enter value: ");
    Int value = sc.nextInt();
    // Shift values to right
    For (int i = size; i > index; i--) {
        Array[i] = array[i - 1];
    }
    Array[index] = value;
    Size++;
    System.out.println("Value inserted successfully.");
}

// 3. Fill array
Public static void fill(Scanner sc) {
    System.out.print("How many values do you want to enter? ");
    Int n = sc.nextInt();
    If (n > 20 - size) {
        System.out.println("Not enough space in array.");
        Return;
    }
    For (int i = 0; i < n; i++) {
        System.out.print("Enter value: ");
        Array[size] = sc.nextInt();
        Size++;
    }
    System.out.println("Array filled successfully.");
}

// 4. Delete last element
Public static void deleteLast() {
    If (size == 0) {
        System.out.println("Array is empty!");
    }
}

```

```

        Return;
    }
    Size--;
    System.out.println("Last value deleted.");
}
// 5. Delete by index
Public static void deleteByIndex(Scanner sc) {
    If (size == 0) {
        System.out.println("Array is empty!");
        Return;
    }
    System.out.print("Enter index to delete: ");
    Int index = sc.nextInt();
    If (index < 0 || index >= size) {
        System.out.println("Invalid index!");
        Return;
    }
    // Shift values to left
    For (int i = index; i < size - 1; i++) {
        Array[i] = array[i + 1];
    }
    Size--;
    System.out.println("Value deleted successfully.");
}
// 6. Display
Public static void display() {
    If (size == 0) {
        System.out.println("Array is empty!");
        Return;
    }

```

```

System.out.println("Array values:");
For (int i = 0; i < size; i++) {
    System.out.println("Index " + i + " = " + array[i]);
}
}

```

// 7. Search

```

Public static void search(Scanner sc) {
    System.out.print("Enter value to search: ");
    Int value = sc.nextInt();
    Boolean found = false;
    For (int i = 0; i < size; i++) {
        If (array[i] == value) {
            System.out.println("Value found at index " + i);
            Found = true;
        }
    }
    If (!found) {
        System.out.println("Value not found.");
    }
}

```

// 8. Get value at index

```

Public static void getValue(Scanner sc) {
    System.out.print("Enter index: ");
    Int index = sc.nextInt();
    If (index < 0 || index >= size) {
        System.out.println("Invalid index!");
        Return;
    }
}

```

```

System.out.println("Value = " + array[index]);

```

```

    }
    // 9. Update value
    Public static void update(Scanner sc) {

        System.out.print("Enter index: ");
        Int index = sc.nextInt();

        If (index < 0 || index >= size) {
            System.out.println("Invalid index!");
            Return;
        }
        System.out.print("Enter new value: ");
        Int value = sc.nextInt();

        Array[index] = value;

        System.out.println("Value updated successfully.");
    }
    // Main
    Public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Int choice;

        Do {

            System.out.println("\n----- ARRAY MENU -----");
            System.out.println("1. Add value");
            System.out.println("2. Insert at index");

```

```
System.out.println("3. Fill array");
System.out.println("4. Delete last element");
System.out.println("5. Delete by index");
System.out.println("6. Display");
System.out.println("7. Search value");
System.out.println("8. Get value at index");
System.out.println("9. Update value");
System.out.println("10. Size");
System.out.println("11. Exit");
```

```
System.out.print("Enter choice: ");
Choice = sc.nextInt();
```

```
Switch (choice) {
```

```
    Case 1:
```

```
        Add(sc);
```

```
        Break;
```

```
    Case 2:
```

```
        Insert(sc);
```

```
        Break;
```

```
    Case 3:
```

```
        Fill(sc);
```

```
        Break;
```

```
    Case 4:
```

```
        deleteLast();
```

```
        break;
```

case 5:

```
    deleteByIndex(sc);  
    break;
```

case 6:

```
    display();  
    break;
```

case 7:

```
    search(sc);  
    break;
```

case 8:

```
    getValue(sc);  
    break;
```

case 9:

```
    update(sc);  
    break;
```

case 10:

```
    System.out.println("Size = " + size);  
    Break;
```

Case 11:

```
    System.out.println("Program ended.");  
    Break;
```

Default:

```

        System.out.println("Invalid choice!");
    }

    } while (choice != 11);

    Sc.close();
}
}

```

➤ **Question # 02:**

```

Import java.util.Scanner;
Public class QueueOperation {
    Static int[] queue = new int[10];
    Static int front = 0;
    Static int rear = -1;
    // Enqueue
    Public static void enqueue(Scanner sc) {
        If (rear == 9) {
            System.out.println("Queue is full!");
            Return;
        }
        System.out.print("Enter value: ");
        Int value = sc.nextInt();

        Rear++;
        Queue[rear] = value;
        System.out.println("Value added successfully.");
    }
    // Dequeue
    Public static void dequeue() {

```

```

    If (rear < front) {
        System.out.println("Queue is empty!");
        Return;
    }
    System.out.println("Removed value: " + queue[front]);

    // Shift values to left
    For (int i = front; i < rear; i++) {
        Queue[i] = queue[i + 1];
    }
    Rear--;
    System.out.println("Value removed successfully.");
}
// Display
Public static void display() {

    If (rear < front) {
        System.out.println("Queue is empty!");
        Return;
    }

    System.out.println("Queue values:");

    For (int i = front; i <= rear; i++) {
        System.out.println(queue[i]);
    }
}

// Size
Public static void size() {

```

```
    If (rear < front) {  
        System.out.println("Queue size = 0");  
    } else {  
        System.out.println("Queue size = " + (rear - front + 1));  
    }  
}
```

// isEmpty

```
Public static void isEmpty() {  
  
    If (rear < front) {  
        System.out.println("Queue is empty.");  
    } else {  
        System.out.println("Queue is not empty.");  
    }  
}
```

// isFull

```
Public static void isFull() {  
  
    If (rear == 9) {  
        System.out.println("Queue is full.");  
    } else {  
        System.out.println("Queue is not full.");  
    }  
}
```

// Main

```
Public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
Int choice;
```

```
Do {
```

```
    System.out.println("\n----- QUEUE MENU -----");
```

```
    System.out.println("1. Enqueue");
```

```
    System.out.println("2. Dequeue");
```

```
    System.out.println("3. Display");
```

```
    System.out.println("4. Size");
```

```
    System.out.println("5. isEmpty");
```

```
    System.out.println("6. isFull");
```

```
    System.out.println("7. Exit");
```

```
    System.out.print("Enter choice: ");
```

```
    Choice = sc.nextInt();
```

```
    Switch (choice) {
```

```
        Case 1:
```

```
            Enqueue(sc);
```

```
            Break;
```

```
        Case 2:
```

```
            Dequeue();
```

```
            Break;
```

```
        Case 3:
```

```
    Display();  
    Break;
```

```
Case 4:  
    Size();  
    Break;
```

```
Case 5:  
    isEmpty();  
    break;
```

```
case 6:  
    isFull();  
    break;
```

```
case 7:  
    System.out.println("Program ended.");  
    Break;
```

```
Default:  
    System.out.println("Invalid choice!");  
}
```

```
} while (choice != 7);
```

```
Sc.close();  
}
```

```
}
```

➤ **Question # 03:**

```

Import java.util.Scanner;
Public class QueueOperation {
    Static int[] queue = new int[10];
    Static int front = 0;
    Static int rear = -1;
    // Enqueue
    Public static void enqueue(Scanner sc) {
        If (rear == 9) {
            System.out.println("Queue is full!");
            Return;
        }
        System.out.print("Enter value: ");
        Int value = sc.nextInt();

        Rear++;
        Queue[rear] = value;
        System.out.println("Value added successfully.");
    }
    // Dequeue
    Public static void dequeue() {
        If (rear < front) {
            System.out.println("Queue is empty!");
            Return;
        }
        System.out.println("Removed value: " + queue[front]);

        // Shift values to left
        For (int i = front; i < rear; i++) {
            Queue[i] = queue[i + 1];
        }
    }
}

```

```

    Rear--;
    System.out.println("Value removed successfully.");
}
// Display
Public static void display() {

    If (rear < front) {
        System.out.println("Queue is empty!");
        Return;
    }

    System.out.println("Queue values:");

    For (int i = front; i <= rear; i++) {
        System.out.println(queue[i]);
    }
}

// Size
Public static void size() {

    If (rear < front) {
        System.out.println("Queue size = 0");
    } else {
        System.out.println("Queue size = " + (rear - front + 1));
    }
}

// isEmpty
Public static void isEmpty() {

```

```
    If (rear < front) {  
        System.out.println("Queue is empty.");  
    } else {  
        System.out.println("Queue is not empty.");  
    }  
}
```

```
// isFull
```

```
Public static void isFull() {
```

```
    If (rear == 9) {  
        System.out.println("Queue is full.");  
    } else {  
        System.out.println("Queue is not full.");  
    }  
}
```

```
// Main
```

```
Public static void main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    Int choice;
```

```
    Do {
```

```
        System.out.println("\n----- QUEUE MENU -----");  
        System.out.println("1. Enqueue");  
        System.out.println("2. Dequeue");
```

```
System.out.println("3. Display");
System.out.println("4. Size");
System.out.println("5. isEmpty");
System.out.println("6. isFull");
System.out.println("7. Exit");
```

```
System.out.print("Enter choice: ");
Choice = sc.nextInt();
```

```
Switch (choice) {
```

```
    Case 1:
```

```
        Enqueue(sc);
        Break;
```

```
    Case 2:
```

```
        Dequeue();
        Break;
```

```
    Case 3:
```

```
        Display();
        Break;
```

```
    Case 4:
```

```
        Size();
        Break;
```

```
    Case 5:
```

```
        isEmpty();
        break;
```

case 6:

isFull();

break;

case 7:

System.out.println("Program ended.");

Break;

Default:

System.out.println("Invalid choice!");

}

} while (choice != 7);

Sc.close();

}

}